

Project Presentation

Topic: AI Powered Resume Analyzer

Presented By:

Aniruddha Adak	- 27600122030
Krishanu Banerjee	- 27600122136
Sayan Chakraborty	- 27600122180
Soni Kumari	- 27600122150

Supervisor: Mrs. Nivedita Das

Department of Computer Science and Engineering



Budge Budge Institute of Technology, Kolkata

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⚠️ PROBLEM

Resumes are rejected before a human ever reads them.



75% ATS rejection rate

Three out of four resumes are filtered out by Applicant Tracking Systems before reaching a recruiter (Harvard Business Review, 2023).



No feedback loop

Candidates rarely learn why they were filtered, leaving them unable to improve subsequent applications.



Keyword-only matching

Existing tools like JobScan optimize for keyword density but ignore multimodal input (screenshots, voice, PDF/DOCX).



Voice practice gap

Mock-interview platforms rely on scheduled human sessions; none offer real-time AI feedback on spoken answers.



OBJECTIVE:

- **Primary Goal:** To bridge the gap between job seekers and their dream roles using generative AI.
- **Improve Resume Quality:** Generate actionable suggestions to enhance resume structure, keywords, formatting, and relevance for better job matching and ATS compatibility.
- **Provide Personalized Career Guidance:** Offer tailored career recommendations, including suitable job roles, skill improvements, and learning paths based on the user's resume.

ABSTRACT:

- ✓ **Platform Overview:** A full-stack web application combining a responsive Next.js frontend and backend.
- ✓ **LLM/AI :** Unlike Uses Gemini and ai for the resume analysis and voice transformation into text, so that user can communicate/chat through any language.
- ✓ **User Experience:** Features a secure dashboard, optimization section and tracking section.
- ✓ **Impact:** The project aims to reduce the rate of rejections of a job candidate in a job interview for an unstructured resume or absence of specific skills.

TECH STACKS:

➤ Frontend:

- **Framework:** Next.js 16, React 19
- **Styling:** Tailwind CSS V4
- **Language:** Typescript

➤ Backend :

- **Primary Framework:** Next.js Server Side
- **API Services :** Gemini api services and servam ai

➤ Database and Authentication:

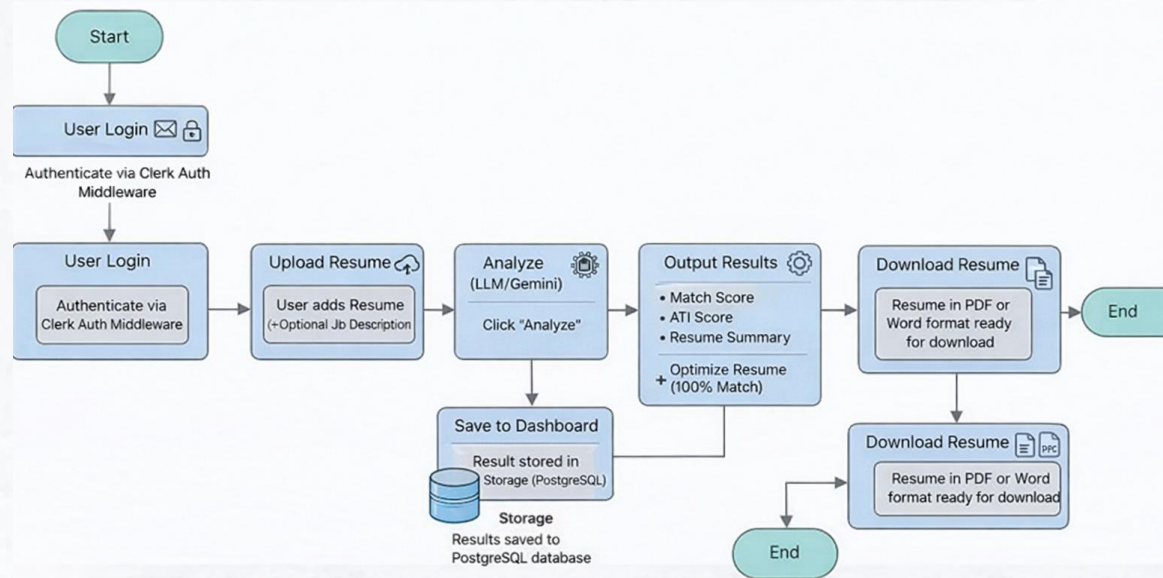
- **Database:** PostgreSQL (vis Prisma ORM)
- **Auth-Middleware :** Clerk

PROPOSED APPROACH:

Web Architecture: We propose a modern full-stack architecture utilizing Next.js for the responsive frontend and server-side rendering, integrated with Clerk Authentication for secure user login and session management. The system enables users to upload resumes and optionally provide job descriptions, ensuring a seamless and secure user experience..

AI-Driven Resume Analysis: Unlike traditional resume screening systems, our approach leverages Large Language Models (LLMs) such as Google Gemini to intelligently analyze resumes. The model evaluates the resume against the job description and generates key insights including Match Score, ATS Compatibility Score, and a structured resume summary, enabling users to understand how well their profile aligns with the job requirements.

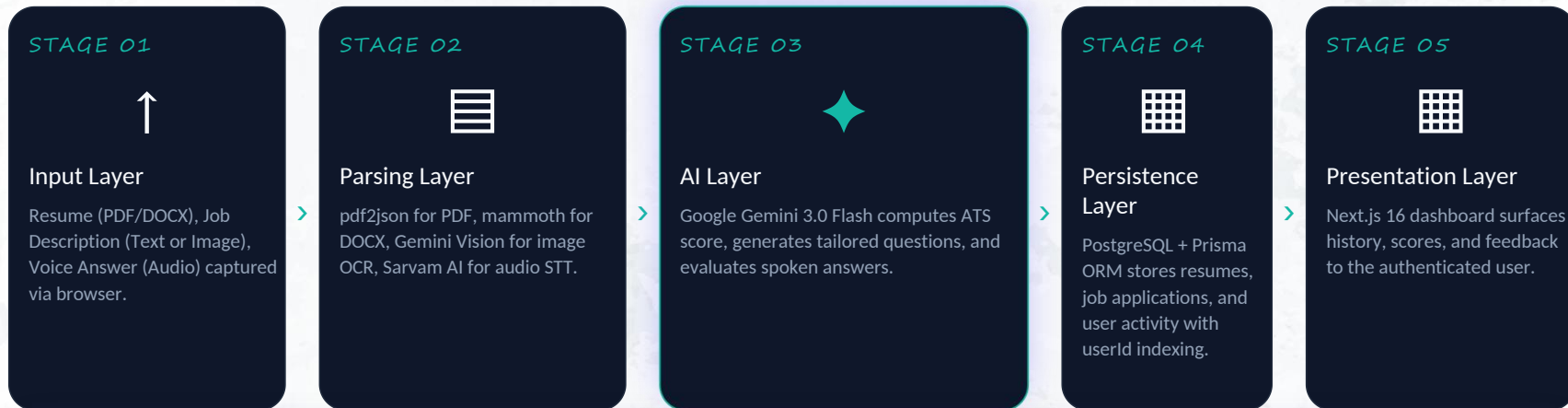
Resume Optimization Engine: To improve hiring success rates, the system provides an AI-powered optimization feature that suggests modifications to increase resume relevance.



SOLUTION ARCHITECTURE

End-to-end multimodal AI pipeline

5 STAGES · 1 LLM



Three input modalities (text, image, voice) converge on a single LLM (*Gemini 3.0 Flash*) for unified reasoning.

UNIQUE FEATURES AND TECHNICAL ADVANTAGES

1.Token-Efficient Resume Processing(PDF → JSON → AI Analysis): Most AI resume analyzers send the entire PDF document directly to the LLM. We first convert the resume into structured JSON using pdf2json. Only relevant fields (Skills, Education, Experience, Projects, Certifications) are sent to Gemini.

- Reduces token consumption significantly.
- Faster response generation.
- Lower API cost.
- Less irrelevant data processed by AI.

2. Structured Resume Understanding: Instead of treating the resume as plain text, the AI receives categorized information.

- More accurate ATS scoring.
- Better skill-gap detection.
- Improved recommendation quality.

3. Skill Gap Analysis: The system identifies:

- Existing skills
- Missing skills
- Recommended skills

4. Explainable ATS Score: Most online ATS checkers simply give a score.

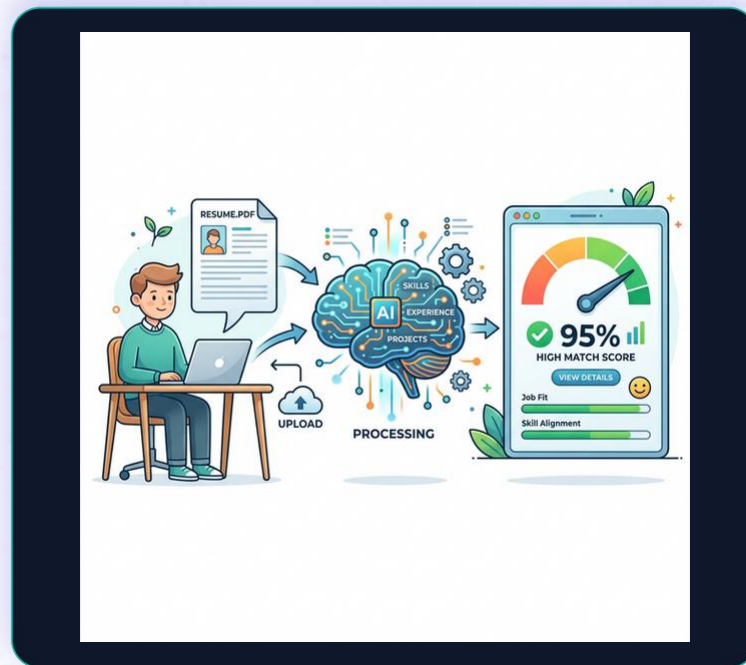
Our system provides:

- ATS Score
- Strengths
- Weaknesses
- Missing Keywords
- Improvement Suggestions

ATS Resume Analysis

Upload resume + job description → get instant ATS score & optimization tips.

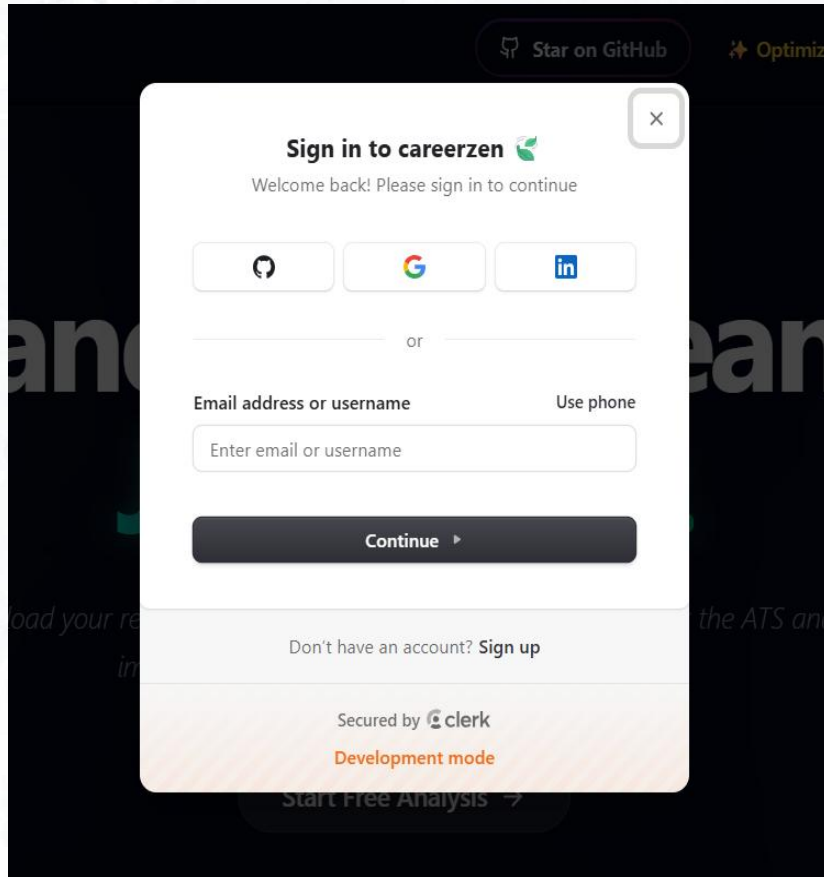
- 1 Drag & drop resume (PDF or DOCX) into the upload zone.
- 2 Paste job description as text OR upload a screenshot of the posting.
- 3 Gemini Vision OCR extracts JD text from uploaded images.
- 4 *pdf2json* (PDF) or *mammoth* (DOCX) extracts resume text with positional sorting.
- 5 Gemini 3.0 Flash compares resume vs JD and computes a 0–100 match score.
- 6 Returns: ATS score, missing keywords, tone critique (optional), and optimization tips.
- 7 Result persisted to PostgreSQL via Prisma and surfaced in the dashboard history.



OVERVIEW OF USER INTERFACE:

The screenshot displays the Careerzen web application interface. At the top, a dark navigation bar contains the Careerzen logo, a 'Star on GitHub' button, and links for 'Optimize', 'Tracker', and 'Dashboard'. A teal banner below the navigation bar reads 'AI-Powered Career Optimization'. The main section features the headline 'Land Your Dream Job Faster.' in large white and teal text, followed by a subtext: 'Upload your resume and get instant, AI-driven feedback to beat the ATS and Impress recruiters. Tailored interview prep included.' A 'Start Free Analysis ->' button is positioned below this text. The lower portion of the interface is a 'Start Analysis' form with the instruction 'Process your resume in seconds.' It includes a 'Resume File *' section with a 'Click to Upload' button (labeled 'PDF or DOCX') and a 'Job Description' section with a text area labeled 'Paste the job description here...' and an 'OPTIONAL' tag. A large teal 'Run Analysis ->' button is at the bottom of the form.

OVERVIEW OF AUTH FORMS:



This is a sign-in form for 'careerzen'. It features a dark background with a white modal box. At the top, it says 'Sign in to careerzen' with a green logo and a close button. Below this is a welcome message: 'Welcome back! Please sign in to continue'. There are three social login buttons: GitHub, Google, and LinkedIn. A separator line with the word 'or' follows. Below the separator, there are two input fields: 'Email address or username' and 'Use phone'. The 'Email address or username' field has a placeholder 'Enter email or username'. A dark 'Continue' button is at the bottom of the modal. At the very bottom of the page, there is a link 'Don't have an account? Sign up' and a footer section that says 'Secured by clerk' and 'Development mode'.

Star on GitHub Optimiz

Sign in to careerzen

Welcome back! Please sign in to continue

GitHub Google LinkedIn

or

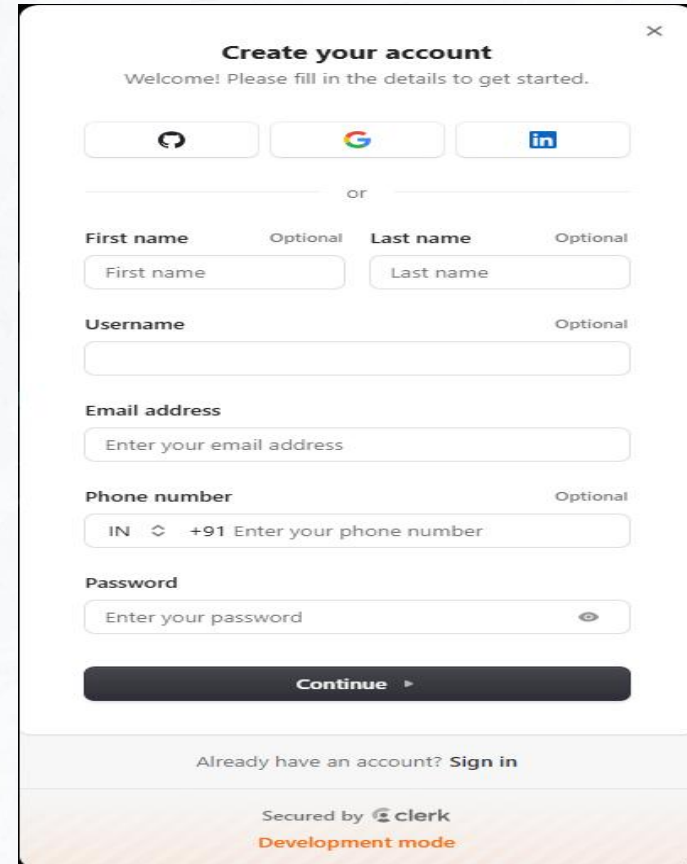
Email address or username Use phone

Enter email or username

Continue ▶

Don't have an account? [Sign up](#)

Secured by clerk
Development mode



This is a 'Create your account' form. It has a white background with a dark header bar. The header says 'Create your account' with a close button and a welcome message: 'Welcome! Please fill in the details to get started.' There are three social login buttons: GitHub, Google, and LinkedIn. A separator line with the word 'or' follows. Below the separator, there are two input fields: 'First name' and 'Last name', both marked as 'Optional'. Below these are 'Username' (Optional) and 'Email address' fields. Below 'Email address' is a 'Phone number' field (Optional) with a dropdown for country code (IN) and a placeholder '+91 Enter your phone number'. Below the phone number is a 'Password' field with a placeholder 'Enter your password' and a toggle for visibility. A dark 'Continue' button is at the bottom of the form. At the very bottom of the page, there is a link 'Already have an account? Sign in' and a footer section that says 'Secured by clerk' and 'Development mode'.

Create your account

Welcome! Please fill in the details to get started.

GitHub Google LinkedIn

or

First name Optional Last name Optional

First name Last name

Username Optional

Email address

Enter your email address

Phone number Optional

IN +91 Enter your phone number

Password

Enter your password

Continue ▶

Already have an account? [Sign in](#)

Secured by clerk
Development mode

OVERVIEW OF DASHBOARD:

 **careerzen**

[Star on GitHub](#) [Optimize](#) [Tracker](#) [Dashboard](#) 

[← Back to Analyzer](#)

Your Library

0 resumes saved

[Application Tracker](#) [Optimize Resume](#) [Analyzer](#)

Your Badges

1/8 earned

**First Steps**
Common

**Getting There**
Common

**Strong Match**
Uncommon

**Elite Resume**
Rare

**Perfect Score**
Legendary

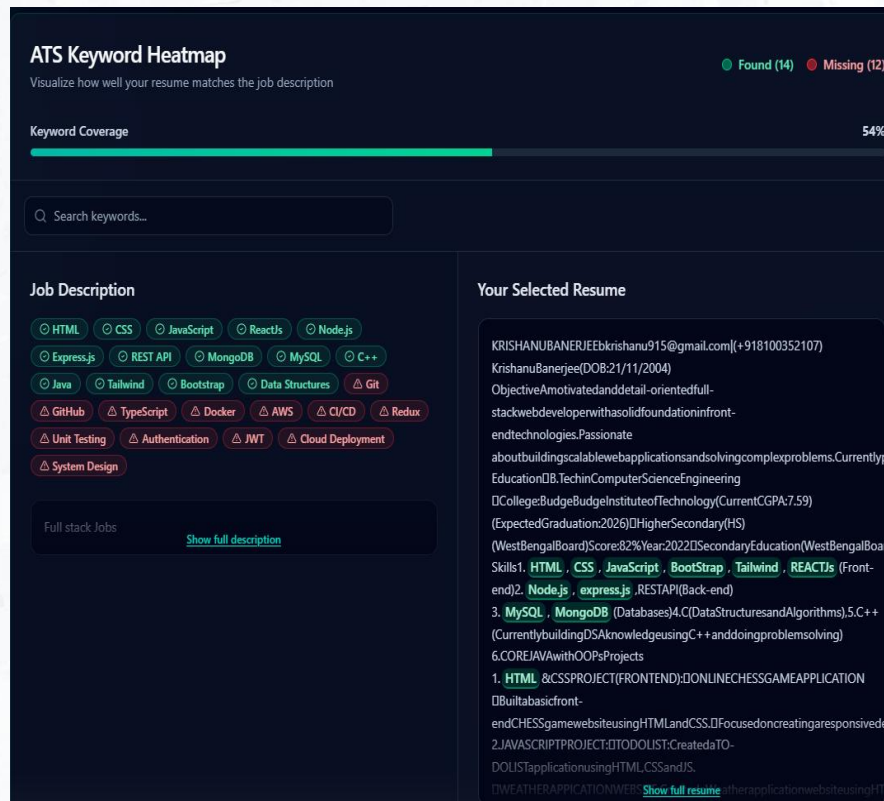
**Optimizer**
Uncommon

**Dedicated**
Rare

**Veteran**
Epic

[Filters](#)

OVERVIEW OF ANALYSIS :



CareerZen demonstrates that multimodal AI can democratize high-quality career preparation



What we built

- ✓ *Production-grade Next.js 16 app* deployed on Vercel with CI/CD via Git pushes.
- ✓ *Integrated 3 AI services* — Gemini 3.0 Flash, Sarvam STT, Gemini Vision — in one workflow.
- ✓ *Open-sourced under MIT* — 73 issues raised, 2 forks, active community engagement.
- ✓ *Validated on real resumes* with $\pm 5\%$ ATS scoring accuracy against a human baseline.



What we learned

- ✓ *Multimodal input* (PDF + image + voice) materially improves user adoption over text-only tools.
- ✓ *Sarvam AI outperforms generic STT* on Indian-English accents — critical for our user base.
- ✓ *Prompt engineering* for structured JSON output from Gemini required iterative discipline.
- ✓ *Denormalized Prisma schema* (no FKs) simplified multi-tenant Clerk userId lookups.



Career prep is no longer a privilege of those who can afford coaches — [open-source AI levels the playing field.](#)



How CareerZen compares to existing tools

4 TOOLS BENCHMARKED

TOOL	ATS SCORE	MULTIMODAL OCR	VOICE STT	AI FEEDBACK	OPEN SOURCE
JobScan	✓	✗	✗	– Limited	✗
ResumeWorded	✓	✗	✗	– Limited	✗
InterviewBuddy	✗	✗	✓ Human	✓ Human	✗
CareerZen (Ours)	✓	✓ Gemini 3.0	✓ Sarvam AI	✓ Gemini	✓ MIT License



CareerZen is the first open-source platform to combine multimodal OCR, voice-based Speech-to-Text, and LLM-driven feedback in a single end-to-end workflow.

CONCLUSION:

- **Bridging The Gap:** The project successfully analyze and fill the gap between the candidate's skillsets and the required skillsets in the production.
- **Guidance:** By this application a job candidate surely can get a guidance for improving his skillsets as recommended.

REFERENCES:

- **Google Gemini 3.0 Flash Preview documentation** -- ai.google.dev/gemini
- **Sarvam AI speech-toText API reference** -- sarvam.ai/docs
- **Next.js 16 App Router documentation** -- nextjs.org/docs
- **Prisma ORM documentation** -- prisma.io/docs
- **Clerk authentication documentation** -- clerk.com/docs
- **Harvard Business Review** -- c “Why your resume gets rejected by ATS”(2023).
- **pdf2json npm package** -- c npmjs.com/package/pdf2json
- **mammoth: DOCX to text** -- c npmjs.com/package/mammoth
- **4, 5, 10**

A soft, abstract watercolor splash in shades of pink, purple, and light blue, with small white speckles, centered on a light cream background.

Thank
You